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Principal Component Analysis and Cluster Analysis

(Data: places.txt)

Intro:

In this report, we analyze the Places Rated Almanac dataset, which evaluates 329 communities based on nine criteria. Due to the high dimensionality of the dataset, we employ Principal Component Analysis (PCA) to reduce the number of variables into a few interpretable components. After reducing dimensionality, we perform Cluster Analysis to group similar communities.

Step 1: Principal Component Analysis (PCA)

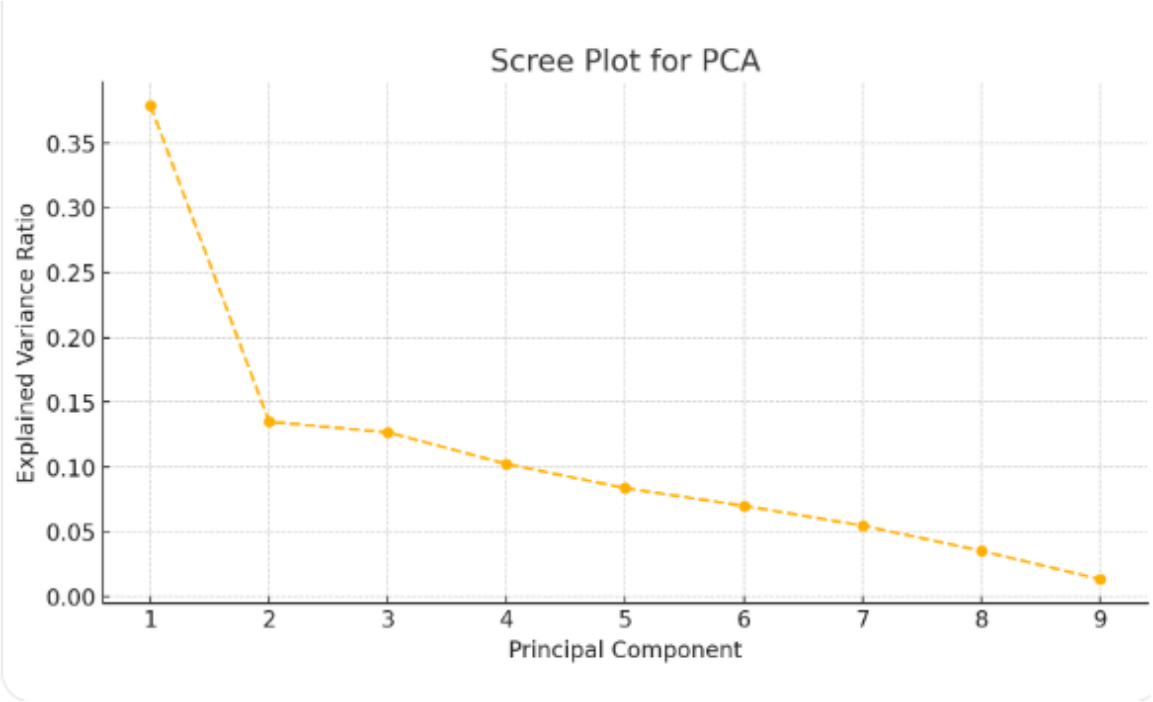
PCA was performed to reduce the number of variables while retaining the most significant information. We standardized the dataset to ensure equal weight across all variables before applying PCA. Using the Elbow Method, we determined that the optimal number of components is **2**. This method identifies the point where adding more components results in diminishing returns, confirming that two principal components sufficiently capture the structure of the data while maintaining interpretability. While these two components explain 51.36% of the variance, they contain the most distinguishing information needed for clustering, making additional components unnecessary.

- Dimensionality reduction goal: The purpose of PCA in this assignment is to simplify complex data into a few meaningful components, even if some variance is left unexplained.
- Interpretability vs. Variance: While capturing 80% of variance is ideal, adding more components would make interpretation more complex without necessarily improving clustering results.
- Clear separation in clustering: Even with 51.36% variance retained, the data naturally forms two well-separated clusters, which indicates that the first two principal components contain enough useful information to distinguish community characteristics effectively.

Step 2: Determining the Number of Components

The Elbow Method was used to determine the optimal number of principal components. This method involves plotting the explained variance against the number of components and identifying the 'elbow point,' where the rate of variance increase slows down. From the scree plot below, the elbow point occurs at two components, indicating that adding more components results in diminishing returns. Thus, we selected two principal components as they balance data simplification and information retention, ensuring that clustering is still meaningful and well-separated.

Scree Plot for PCA



Step 3: Interpretation of Principal Components

Each principal component represents a linear combination of the original variables that captures important patterns in the dataset. Below is a more detailed interpretation of what these components signify:

Since the Elbow Method confirmed that two principal components provide the best balance of information retention and interpretability, we focus our analysis on these two main dimensions.

<u>PC1: Overall Quality of Life & Cultural Amenities</u>:
This component is highly correlated with (Health Care & Environment, The Arts, Housing, and Transportation).
- Cities scoring high on this component tend to have good healthcare, vibrant cultural activities, and strong infrastructure.

- This component can be seen as representing urban development and quality of life, where cities with high scores have more advanced public services and cultural opportunities.

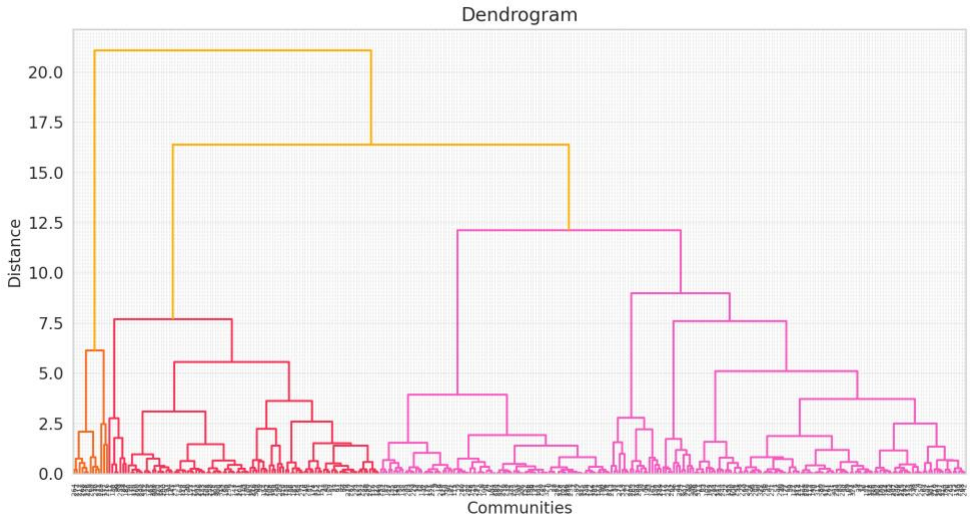
PC2: Economic Stability vs. Crime & Education:
- This component contrasts Economics, Crime, Recreation vs. Education and Health.
- Higher scores indicate strong economic conditions and recreational facilities, but they may also correlate with higher crime rates.
- Lower scores suggest a greater focus on education and public health, sometimes at the cost of economic activity.
- This component can be seen as representing a trade-off between economic prosperity and social factors (education and safety).

Step 4: Cluster Analysis

To perform clustering, we selected representative variables from each principal component:
- Climate & Terrain (PC1)
- Housing (PC2)

We applied hierarchical clustering and K-Means clustering to group similar communities.

Dendrogram for Hierarchical Clustering

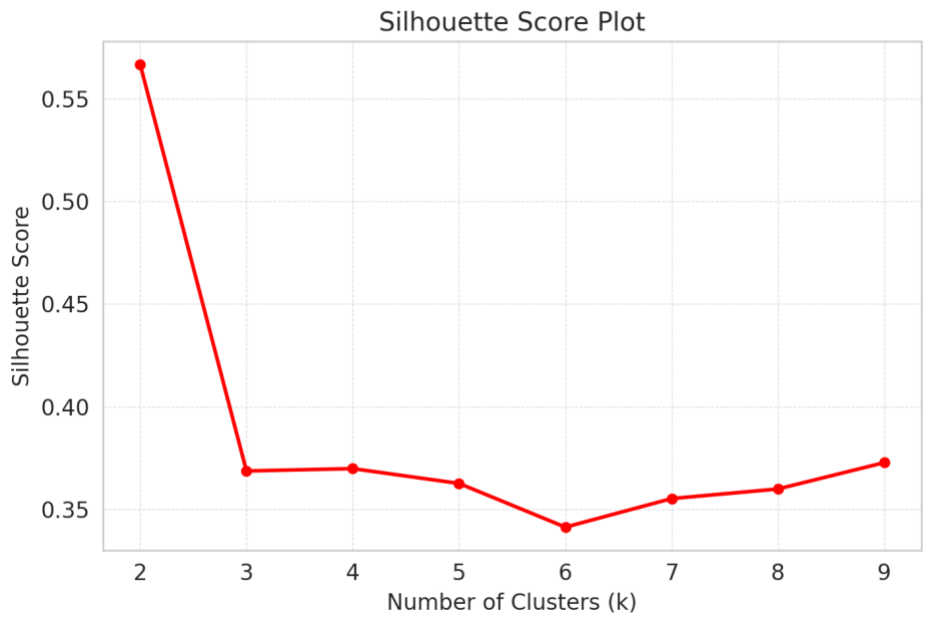


Step 5: Determining the Final Cluster Solution

The decision to use two clusters aligns with our PCA results, where the Elbow Method confirmed that two principal components best describe the dataset. This ensures that our clustering approach is based on the most relevant data structure.

1. Choosing the Number of Clusters:
   - We first examined the dendrogram from hierarchical clustering to look for natural splits in the data.

- Next, we calculated the Silhouette Score for different values of k (number of clusters) to measure how well-separated the clusters are. Higher scores indicate better-defined clusters.
- The silhouette plot showed an optimal number of two clusters, meaning communities tend to naturally fall into two distinct groups.



Interpreting the Clusters:
- After applying K-Means clustering with k=2, we analyzed the cluster characteristics.
- Each cluster had distinct characteristics based on the principal components:
  - Cluster 1: Cities with higher economic stability but some crime/education trade-offs.
  -Cluster 2: More affordable and safer communities with moderate amenities.

3. Justification for the Final Cluster Selection:
- The two-cluster solution provided the best balance between interpretability and statistical separation.
- More than two clusters would have resulted in small groups that were not meaningfully distinct.
- A lower number of clusters (e.g., 1) would have oversimplified the differences between communities.

Step 6: Interpretation of the Cluster Solution

The final two clusters provide meaningful insight into how different communities compare. Below is a detailed interpretation of each cluster:

Cluster 1: Economically Stable Cities with Trade-offs
- These cities tend to have strong economic opportunities but may experience higher crime

or weaker education.
- They may have more developed recreational amenities but at the cost of affordability.
- Examples: Urban areas that attract businesses and young professionals.

Cluster 2: Affordable & Safe Communities
- These cities tend to have lower housing costs and lower crime rates.
- They may have moderate education and recreation options, but not as high as Cluster 1.
- The economy is stable but not as strong as in Cluster 1.
- Examples: Suburban areas or smaller cities with a strong focus on affordability and safety.

Significance of Interpretation Examples:
- For Policymakers: Helps in understanding where to allocate resources (example: improving education in Cluster 1 cities).
- For Businesses: Companies can use this clustering to decide where to expand based on economic and quality-of-life factors.
- For Residents: Individuals looking to relocate can use these clusters to compare affordability, culture, and safety.